

An Overview of SDMA in Communications Systems

Iclia Villordo-Jiménez, Ignacio E. Zaldivar-Huerta

Department of Electronics
INAOE

Calle Luis Enrique Erro No. 1, C.P. 72840
Tonantzintla, Pue., Mex.
Email: {iclia, zaldivar}@inaoep.mx

G. M. Galvan-Tejada

Communications Section. Department of Electrical
Engineering. CINVESTAV-IP.N.
Av. IPN 2508 Col. San Pedro Zacatenco C.P. 07360
México, D.F.
Email: ggalvan@cinvestav.mx

Abstract—Space Division Multiple Access (SDMA) is a technology that permits the increase of capacity and performance of communications systems. In this paper, the main characteristics, advantages, disadvantages and the effects of the use of SDMA on a communication system are showed. Additionally an overview of smart antenna technology is given.

I. INTRODUCTION

Nowadays, mobile communication systems are increasing the number of users that demands new services, such as audio and video of high quality, e-mail, videoconference and Internet, among others [1, 2]. In this sense, researchers are currently attempting to satisfy these requirements for future generations of mobile communication systems. For this reason, is necessary the use of new technology that permits the increase of capacity of the radio communication networks.

A proposal to achieve the previous goals is the use of Space Division Multiple Access (SDMA), which is based on smart antenna technology [3]. This technology has its origins in satellite communications and has been developed in radar systems [4]. SDMA technique allows the simultaneous use of a radio channel by multiple users within the same cell considering their spatial position. By consequence, a smart antenna is required to dynamically optimize the transmission and reception of radiation patterns in response to the surrounding environment. In this sense, the spectral efficiency and the capacity of the system are limited by the environmental interference [1-4]. For this reason, the main characteristics of SDMA will be analyzed in the next sections.

This paper is organized as follows. Section II shows an overview of smart antennas. Section III describes the advantages and disadvantages of SDMA technique. The trends of SDMA are summarized in Section IV and finally, Section V gives the conclusions.

II. SMART ANTENNA TECHNOLOGY

A smart antenna consists of an antenna array and a digital signal processing stage as shown in Fig. 1, where the antenna array is controlled by the digital signal processing stage in response to the radio frequency environment [4].

In this sense, the radiation pattern of each element of the array is automatically adapted by the smart antenna to identify the spatial position of a desired user and cancels the co-channel interference from other users within the cell and adjacent cells.

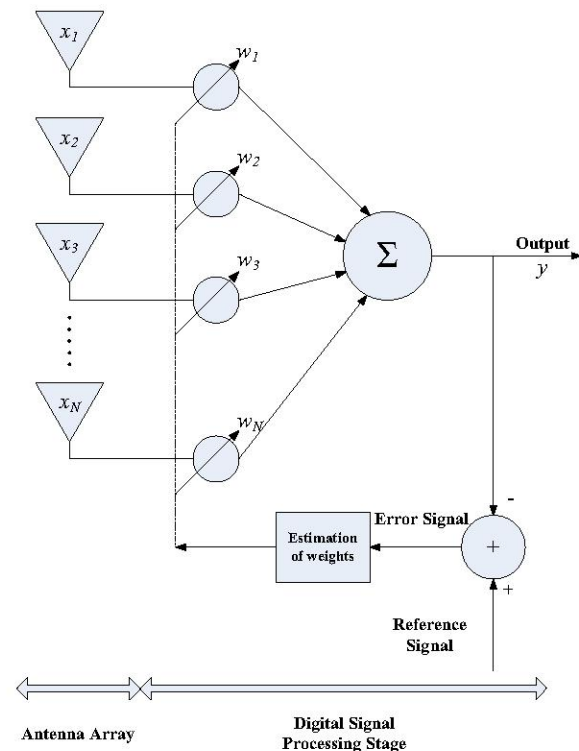


Figure 1. Smart antenna system.

The above is achieved by reducing the side lobes in the directions of interferences [2]. The signals at the output of each element are processed considering the inherent spatial information of the users. These signals are multiplied by certain weights, and if they are selected appropriately, the radiation pattern of the array can be controlled to improve the reception of the desired user and reject the interference users [4]. It is important to remark that the array is not smart by itself, and is necessary the digital signal processing stage because provides the adaptive feature.

Different types of arrays can be used for a smart antenna system, being the most used the uniform linear and circular arrays [4, 5]. A uniform linear array is composed by N elements aligned and with the same interelement spacing d , as can be seen in Fig. 2 [4]. If a wave front arrives at the array with an angle θ_0 (corresponding to the desired user direction), the array factor is defined as [4]:

$$F(\theta) = \sum_{n=0}^{N-1} A_n e^{j(\kappa n d \sin \theta + n\alpha)} \quad (1)$$

Where A_n is the signal amplitude, $\kappa=2\pi/\lambda$, λ is the wavelength of interest, α is the phase delay between adjacent elements and θ is the azimuthal angle. If $\alpha = -\kappa d \sin \theta_0$, the maximum response of $F(\theta)$ will be in θ_0 , that is, the main lobe is pointed to the desired user.

On the other hand, a uniform circular array is composed by N elements in a circle of radius R , as depicted in Fig. 3, where $\beta=2\pi/N$, is the relative angle between adjacent elements. The elevation angle φ is measured from z axis. The azimuthal angle θ is measured in the clockwise sense.

For this case, the array factor is defined as [4]:

$$F(\varphi, \theta) = \sum_{n=0}^{N-1} A_n e^{j[\alpha_n - \kappa R \sin \varphi \cos(\theta - \theta_n)]} \quad (2)$$

Where $\theta_n=2\pi n/N$ and α_n is the phase delay of the n th element. If $\alpha_n = \kappa R \sin \varphi_0 \cos(\theta_0 - \theta_n)$, the main lobe will be pointing to (φ_0, θ_0) in the space.

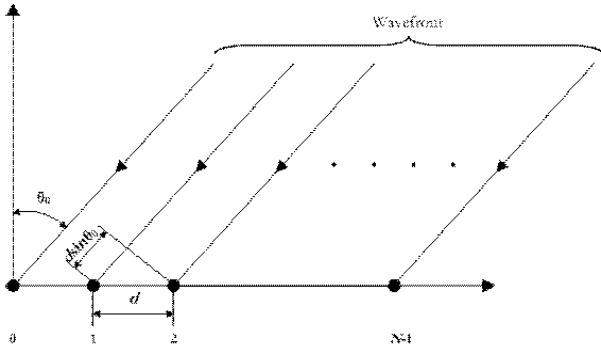


Figure 2. Uniform linear array.

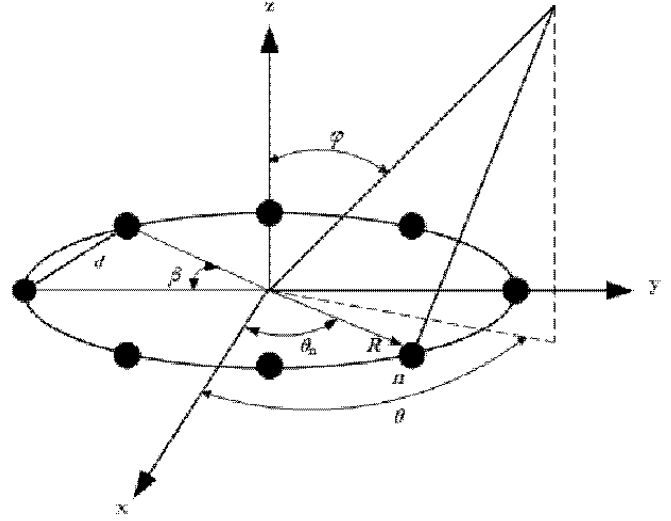


Figure 3. Uniform circular array.

In some references, the radius of this array is selected taking into account two criteria [4-6]. The first one is to consider the linear interelement spacing, where the optimum value is 0.5λ according to a linear distance. Under this consideration the radius is computed as [6]:

$$R = \frac{\lambda}{4 \left| \sin \frac{\pi}{N} \right|} \quad (3)$$

The second criterion is based on the number of modes that could be excited by the array, and now the radius is computed as [5]:

$$R < \frac{\lambda N}{4\pi} \quad (4)$$

Fig. 4 shows two array factors obtained using (1) and (2) and considering an array of 12 elements. For the uniform linear array the radiation pattern is altered when θ takes values near to 0 or 180 degrees, causing ambiguities in the desired direction. By this reason a linear array has coverage of only 180 degrees. On the other hand, the uniform circular array has coverage of 360 degrees without any problem.

An important characteristic of a smart antenna is the selection of the algorithm for the digital signal processing stage. A good selection of the algorithm permits that the weights of the array can be adjusted to steer the main lobe to the desired user and reject the interference caused by co-channel users located in the same cell and in other cells. By consequence, the selection of the algorithm is very important because it determines the velocity, convergence and complexity of the hardware. In this context, the adaptive algorithms offer an attractive performance, being the most utilized: LMS, RLS, Direct Sample Covariance Matrix Inversion and Neural Networks [2, 3].

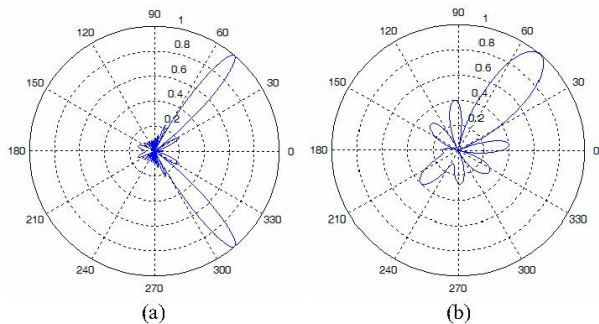


Figure 4. Array factor of the arrays, (a) Uniform linear array, (b) Uniform circular array.

III. SDMA

The characteristics of SDMA are very peculiar in comparison to others schemes of multiple access, such as FDMA, TDMA and CDMA. The most important characteristic is that SDMA allows the simultaneous reuse of any conventional channel (frequency band, time slot or code) by multiple users of the same cell. This is achieved using the spatial position of the users relative to the base station as is illustrated in Fig. 5 [1, 2]. The second most important characteristic is that SDMA can not be implemented by itself in a system; SDMA must be superimposed on other multiple access technique. In this sense, the system increases its spectrum efficiency and does not require an increment of the bandwidth to serve the new users [3].

A factor that limits the increase of capacity of a communication system is the co-channel interference. This effect is product of the reuse of the available channels within the system. With the use of SDMA, this interference is reduced significantly employing spatial filtering techniques by means of smart antenna technology [2]. Also, the transmission efficiency is increased because the covering area is larger [4]. In addition, the power transmission is reduced and the power consume in the mobile could be lowered, and this improves the power control [3].

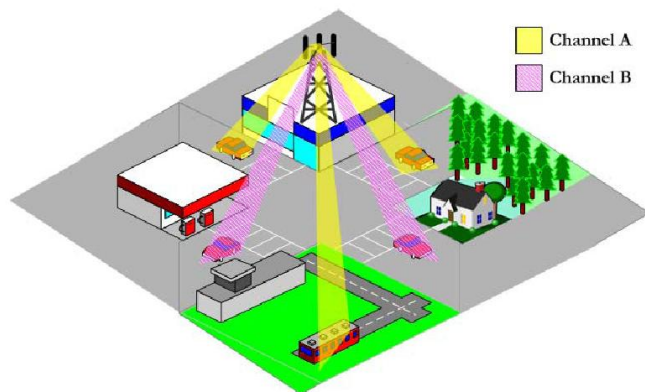


Figure 5. Channel spatial reuse in SDMA.

A system with SDMA is robust against the multipath fading due to the number of multipath signals is reduced and the spatial diversity inherent to the antenna array provides an efficient countermeasure [3, 4]. Independently that SDMA can be used with fixed or dynamic channel allocation, other characteristic in a system with SDMA is the intracell transference [6]. This is, if the signal quality of the users that share the same channel decreases, one or more users must be transferred to other channel or blocked if there are not available channels. SDMA needs some particular requirements and criteria for the radio resource management as can be shown in [9, 10].

Although the specific conditions of the system (e.g. the multipath environment, channel allocation, mobility, number of users, QoS of the users, etc.), it is clear that implementation of the system requires a complex network due to the digital signal processing algorithm selected. In general, the transmission and reception spatially selective of the signals, gives an important increase of capacity, coverage and quality in a communication system [8-10]. In this context, different authors show that a system using SDMA has better performance (an increase of capacity and signal quality) than a system without SDMA [2, 9, 10].

The total cost of the SDMA system includes the hardware for the control of the smart antenna, the construction of the array, the physical implementation of the smart antenna, the electrical power amplifiers, and the control circuits, among others. For this reason, the implementation of this technique is expensive [1, 2].

IV. TRENDS OF SDMA

Commercially, it has been projected that there will be more than one million smart antenna base stations in use for Personal Communications Systems (PCS)/3G in this year [8]. According to [2], until 2004 smart antenna technology was used in one of every 10 base stations in the world and by 2008 is predicted an increase of 60 percent of the use of this technology. Until 2003, Arraycomm Inc. has installed over 100,000 smart base stations in the world, where 60,000 of them are in Japan [7, 8]. The company affirms that the cost of the equipment is only 15 percent higher than conventional equipment. Ericsson Inc. has showed experimentally the possibility of use one smart base station for every 2.5 conventional base stations. Taking into account that telecommunication market is growing every day; in 2005 Lucent Inc. has opened a new line of devices with application in 3G base stations. In this sense, by 2001, Metawave Inc. reported sales over US\$100 million in smart antenna products and services only in US cellular markets [8]. The financial impact of the use of smart antenna technology in future wireless systems was reported in [11] for UMTS and cdma2000. These results permit to predict that the key to make sure the financial viability of operator's business is smart antenna technology. Thus, with the increase of smart antenna technology in the currents and future communications systems, is more feasible the use of SDMA in those systems.

V. CONCLUSIONS

In this paper, a revision of the smart antenna technology and a study of SDMA has been carried out. An important feature is that this multiple access scheme can improve the capacity of communication systems and increase the spectrum efficiency. Also, SDMA improves the quality of service of the communication systems. However, there are many details that still must be planned for implement it efficiently.

REFERENCES

- [1] T. Kaiser, "When will smart antennas be ready for the market? Part I", IEEE Signal Processing Magazine, Vol. 22, Issue 2, pp. 87-92, March 2005.
- [2] A. Alexiou and M. Haardt, "Smart antenna technologies for future wireless systems: trends and challenges", IEEE Communications Magazine, Vol. 42, Issue 9, pp. 90-97, September 2004.
- [3] H. Jin and A. Acampora, "A reservation-based media access control (MAC) protocol design for cellular systems using smart antennas-Part I. Flat fading", IEEE Transactions on Wireless Communications, Vol. 4, No. 2, pp. 792-801, March 2005.
- [4] C. A. Balanis, *Antenna Theory: Analysis and design*, 3rd Edition, Wiley, 2005.
- [5] P. Ioannides and C. A. Balanis, "Uniform circular arrays for smart antennas", IEEE Antennas and Propagation Society Int. Symposium, Vol. 3, pp. 2796-2799, June 2004.
- [6] S. Durrani and M. E. Bialkowski, "Effect of mutual coupling on the interference rejection capabilities of linear and circular arrays in CDMA systems", IEEE Trans. on Antennas and Propagation, Vol. 52, No. 4, pp. 1130-1134, April 2004.
- [7] <http://www.visantstrategies.com>
- [8] <http://www.touchbriefings.com>
- [9] F. Piolini and A. Rolando, "Smart channel-assignment algorithm for SDMA systems", IEEE Trans. on Microwave Theory and Techniques, Vol. 47, pp. 693-699, 1999.
- [10] I. Villordo-Jimenez and G. M. Galvan-Tejada, "Evaluation of spatial channel allocation algorithms for space division multiple access", 1st Int. Conf. on Electrical and Electronics Engineering ICEEE-2004, pp. 108-112, June 2004.
- [11] D. Avidor, "On the financial impact of capacity enhancing technologies to wireless operators", IEEE Trans. on Wireless Commun., Vol. 10, Issue 4, pp. 62-65, August 2003.